

PROJECT REPORT OF CO-OP

Project at Industry (Module-2)

ON

Hybrid CNN-LSTM Framework for Spatial-Temporal Air Pollution Forecasting in Smart Cities

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DECLARATION

We hereby certify that the work presented in the project report entitled "Hybrid CNN-LSTM Framework for Spatial-Temporal Air Pollution Forecasting in Smart Cities" in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering (Computer Science and Engineering), submitted in the Department of Computer Science and Engineering, Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of Ms. Isha Kansal. The matter presented in this report has not been submitted to any other university/institute for the award of any degree.

Place: Rajpura

This is to certify that the above statement made by the candidates is correct to the best of my knowledge and belief.

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ABSTRACT

This project presents a Hybrid CNN-LSTM deep learning model for forecasting air pollution levels — specifically PM_{2.5} concentration — across multiple Indian cities. The model combines Convolutional Neural Networks (CNN) for spatial feature extraction with Long Short-Term Memory (LSTM) networks for temporal sequence learning, enabling accurate multi-step air quality forecasting.

The model was trained and evaluated on the publicly available Air Pollution in India dataset (Kaggle/CPCB), comprising six pollutant variables and three meteorological covariates. The proposed architecture outperformed baseline models — ARIMA, SVR, standalone CNN, and standalone LSTM — achieving an RMSE of 12.47, MAE of 9.83, and R^2 of 0.941 for 1-hour PM_{2.5} forecasting.

Keywords: *Air Quality Forecasting, CNN, LSTM, Deep Learning, PM_{2.5}, Smart Cities*

1. Introduction

Air pollution is one of the most pressing environmental and public health challenges of the 21st century. Rapid urbanization, industrial growth, and increasing vehicular traffic in Indian cities have led to dangerous levels of particulate matter and gaseous pollutants. Accurate and timely air quality forecasting is essential for enabling early warnings, informing policy decisions, and protecting public health.

Traditional statistical models such as ARIMA fail to capture the non-linear and multi-variable nature of pollution data. Classical machine learning approaches like Support Vector Regression (SVR) improve upon this but cannot model the sequential temporal dependencies inherent in air quality time series. Deep learning, particularly LSTM-based models, handles temporal data well but lacks the ability to extract spatial cross-variable features simultaneously.

This project proposes a Hybrid CNN-LSTM architecture that addresses these limitations by combining the feature extraction strength of Convolutional Neural Networks with the temporal modeling capability of LSTM networks. The system operates on multivariate time-series data — pollutant concentrations and meteorological parameters — to forecast PM2.5 levels across multiple time horizons.

1.1 Objectives

- To design and implement a Hybrid CNN-LSTM model for multi-step PM2.5 forecasting.
- To develop a robust preprocessing pipeline for handling missing values, outliers, and normalization.
- To evaluate the proposed model against standard baselines (ARIMA, SVR, CNN, LSTM).
- To analyse feature importance and identify key drivers of air pollution levels.

2. Methodology

The project followed a structured pipeline from raw data acquisition to final model evaluation. The overall workflow is described below.

2.1 Data Collection

The Air Pollution in India dataset was sourced from Kaggle, aggregated from real-time monitoring stations of the Central Pollution Control Board (CPCB) across multiple Indian cities. It contains six pollutant variables (PM2.5, PM10, NO2, SO2, CO, O3) and three meteorological covariates (temperature, humidity, wind speed), forming a 9-feature multivariate input space. The target variable for forecasting is PM2.5.

2.2 Data Preprocessing

- Missing Values: Linear interpolation for gaps up to 3 time steps; KNN imputation ($k=5$) for larger gaps.
- Outlier Detection: IQR criterion (threshold: $Q3 + 3 \times IQR$). Sensor-fault outliers replaced with temporal median; genuine pollution spikes retained.

- Normalization: Min-max scaling to $[0, 1]$ using training-set parameters only, preventing data leakage.

2.3 Sequence Construction

A sliding window of $T = 24$ hours was used to create input sequences of shape $(24, 9)$. Prediction horizons of $H = 1, 6$, and 24 hours were defined. The window slides one step at a time across the dataset.

2.4 Model Architecture

The Hybrid CNN-LSTM model consists of the following stages:

- CNN Block: Two 1D convolutional layers (64 and 128 filters, kernel size 3, ReLU activation) with batch normalization and MaxPooling, reducing temporal dimension by half.
- LSTM Block: Two stacked LSTM layers (128 and 64 units) with dropout (rate 0.3) on recurrent connections for temporal learning.
- Regression Head: Two dense layers (32 and 16 units, ReLU) followed by a linear output neuron for PM2.5 prediction.

2.5 Training Configuration

- Optimizer: Adam with learning rate 1×10^{-3} and exponential decay (0.95 every 10 epochs).
- Loss Function: Mean Squared Error (MSE).
- Epochs: Up to 200 with early stopping (patience = 20).
- Batch Size: 64.
- Data Split: 70% training / 15% validation / 15% testing (temporal order preserved).

3. Tools and Technologies

Tool / Library	Version	Purpose
Python	3.9	Core programming language
TensorFlow / Keras	2.10	Deep learning model development
NumPy	1.23	Numerical computation and array handling
Pandas	1.5	Data manipulation and time-series processing
Scikit-learn	1.1	SVR baseline, normalization, metrics
Matplotlib / Seaborn	Latest	Data visualization and result plotting
Jupyter Notebook	Latest	Interactive development and experimentation
Kaggle (Dataset)	—	Source of Air Pollution in India dataset

3.1 Hardware Environment

- GPU: NVIDIA Tesla T4 (16 GB VRAM) — for accelerated model training.
- CPU: Intel Xeon 2.2 GHz.
- RAM: 32 GB.
- OS: Ubuntu 20.04 LTS.

4. Implementation

4.1 Dataset Preparation

The raw dataset was loaded using Pandas. Timestamp columns were parsed and set as the index. Missing values were handled using linear interpolation and KNN imputation. Outliers identified via IQR were either replaced or retained based on quality flags. All features were then scaled to $[0, 1]$ using min-max normalization computed on the training split.

4.2 Sliding Window Sequences

Overlapping windows of 24 timesteps were extracted to form input tensors of shape (24, 9). Corresponding output labels were the PM2.5 values at +1, +6, and +24 hours ahead. The dataset was split into training (70%), validation (15%), and test (15%) sets maintaining chronological order.

4.3 Model Construction (Keras)

The model was built using the Keras Sequential API. The CNN block was constructed with two Conv1D layers, followed by BatchNormalization, MaxPooling1D, and Dropout layers. The output was then passed to two stacked LSTM layers with inter-layer dropout. Finally, Dense layers reduced the output to a single PM2.5 prediction. For multi-step forecasting ($H = 6, 24$), the output layer was adjusted to H neurons.

4.4 Training and Validation

The model was compiled with Adam optimizer and MSE loss. Training used EarlyStopping on validation loss with patience of 20 epochs and ModelCheckpoint to save the best weights. Training and validation loss curves were monitored to detect overfitting. The best model weights were restored after early stopping.

4.5 Baseline Comparisons

Four baseline models were trained and evaluated on the same data splits:

- ARIMA: Classical statistical model fit on univariate PM2.5 series.
- SVR: Support Vector Regression with RBF kernel on the same multivariate feature set.
- Standalone CNN: The CNN block only, without LSTM.
- Standalone LSTM: The LSTM block only, without CNN preprocessing.

5. Major Findings / Outcomes / Results

5.1 Performance Comparison

Table 3 summarises the forecasting performance across all models and time horizons. All values are averaged over five runs with different random seeds.

Table 3: Forecasting Performance Comparison Across Models and Horizons

Model	RMSE (1h)	MAE (1h)	R ² (1h)	RMSE (6h)	RMSE (24h)	R ² (24h)
ARIMA	31.82	25.14	0.731	39.54	58.21	0.612
SVR	24.57	19.83	0.812	31.87	49.63	0.701
CNN only	19.31	15.42	0.891	26.54	41.27	0.798
LSTM only	16.88	13.21	0.921	23.87	38.14	0.832
CNN-LSTM (Proposed)	12.47	9.83	0.941	18.93	29.45	0.912

5.2 Key Findings

- The Hybrid CNN-LSTM achieved the best performance across all horizons, with an R² of 0.941 for 1-hour forecasting — meaning the model explains 94.1% of variance in observed PM_{2.5} values.
- Compared to standalone LSTM, the hybrid model reduced RMSE by 26%, confirming the benefit of CNN-based spatial feature preprocessing.
- Performance degraded gracefully with increasing forecast horizon — RMSE rose from 12.47 (1h) to 29.45 (24h) — indicating stable long-range forecasting capability.
- ARIMA performed the worst, confirming the limitations of linear statistical models on non-linear pollution data.

5.3 Feature Importance

- PM₁₀ was the most influential predictor of PM_{2.5}, consistent with shared emission sources (vehicular and industrial).

- Wind speed ranked second — higher wind speeds accelerate pollutant dispersion and reduce PM_{2.5} levels.
- Temperature and humidity were important during summer (ozone formation) and monsoon (particle hygroscopic growth) seasons respectively.
- NO₂ and CO served as reliable tracers of vehicular combustion emissions.

5.4 Ablation Study

Removing individual components from the full model showed: batch normalization contributed a 4.2% RMSE reduction, dropout regularization contributed 3.1%, the second CNN layer contributed 2.8%, and stacked LSTM layers contributed 5.7%. Each component meaningfully improved overall performance.

6. Conclusion and Future Scope

6.1 Conclusion

This project successfully developed and evaluated a Hybrid CNN-LSTM framework for air quality forecasting. The proposed architecture effectively combines spatial feature extraction (CNN) and temporal sequence modeling (LSTM) to forecast PM_{2.5} concentrations across 1-hour, 6-hour, and 24-hour horizons. The model demonstrated superior accuracy over all four baselines on the real-world Indian air quality dataset, achieving an RMSE of 12.47, MAE of 9.83, and R^2 of 0.941.

The project demonstrates that deep learning, when thoughtfully designed, is a powerful and scalable tool for environmental monitoring in smart cities. The framework can support real-time air quality alert systems, traffic management decisions, and public health advisories.

6.2 Future Scope

- Attention Mechanisms: Adding multi-head self-attention (Transformer-style) layers to improve selective weighting of time steps, especially for longer forecasting horizons.

- Graph Neural Networks: Encoding spatial relationships between geographically distributed sensor stations for true spatial-temporal modeling.
- Multi-Pollutant Forecasting: Extending the output head to simultaneously predict all six pollutants for a comprehensive Air Quality Index.
- Edge Deployment: Optimising the model for deployment on IoT edge devices (e.g., Raspberry Pi, NVIDIA Jetson Nano) for real-time city-level monitoring.
- Satellite Data Integration: Incorporating Sentinel-5P or MODIS AOD satellite observations to augment ground sensor data, especially in data-sparse regions.
- Uncertainty Quantification: Using Bayesian deep learning or Monte Carlo Dropout to generate confidence intervals around predictions for risk-aware decision-making.

7. References

- [1] World Health Organization. (2021). Global Air Quality Guidelines. WHO Press.
- [2] IQAir. (2023). World Air Quality Report. IQAir Group.
- [3] Bai, L. et al. (2018). Air pollution forecasts: An overview. *Int. J. Environ. Res. Public Health*, 15(4), 780.
- [4] Hochreiter, S. & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780.
- [5] Kumar, A. & Jain, V. (2021). Predicting AQI using hybrid ML and DL models. *Environ. Sci. Pollut. Res.*, 28(29).
- [6] Huang, C.-J. & Kuo, P.-H. (2018). A deep CNN-LSTM model for PM_{2.5} forecasting in smart cities. *Sensors*, 18(7), 2220.
- [7] Zhao, J. et al. (2019). LSTM-FC neural network for PM_{2.5} concentration prediction. *Chemosphere*, 220, 486–492.
- [8] Ong, B. T. et al. (2016). Dynamically pre-trained deep RNNs for predicting PM_{2.5}. *Neural Comput. Appl.*, 27(6).
- [9] Li, X. et al. (2017). LSTM neural network for air pollutant concentration predictions. *Environ. Pollution*, 231.
- [10] Liang, Y. et al. (2018). GeoMAN: Multi-level attention networks for geosensory time series. *IJCAI 2018*.
- [11] Yi, X. et al. (2018). Deep distributed fusion network for air quality prediction. *ACM SIGKDD 2018*.
- [12] Xu, Y. et al. (2018). ML techniques for estimating ground-level PM_{2.5}. *Environ. Pollution*, 242, 1417–1426.
- [13] Chen, G. et al. (2018). ML method to estimate PM_{2.5} across China. *Sci. Total Environ.*, 636, 52–60.
- [14] Kingma, D. P. & Ba, J. (2015). Adam: A method for stochastic optimization. *ICLR 2015*.
- [15] LeCun, Y., Bengio, Y. & Hinton, G. (2015). Deep learning. *Nature*, 521(7553), 436–444.

8. Appendices

Appendix A: Dataset Summary

Attribute	Detail
Source	Kaggle / CPCB NAMP
Cities Covered	Multiple Indian cities
Pollutant Features	PM2.5, PM10, NO2, SO2, CO, O3
Meteorological Features	Temperature, Relative Humidity, Wind Speed
Total Features	9
Missing Value Rate	~8.3%
Target Variable	PM2.5 ($\mu\text{g}/\text{m}^3$)
Train / Val / Test Split	70% / 15% / 15%

Appendix B: Model Hyperparameter Configuration

Hyperparameter	Value
CNN Filters (Layer 1 / Layer 2)	64 / 128
CNN Kernel Size	3
Pooling Size (MaxPool)	2
CNN Dropout Rate	0.2
LSTM Units (Layer 1 / Layer 2)	128 / 64
LSTM Dropout Rate	0.3
Dense Units (Layer 1 / Layer 2)	32 / 16
Dense Dropout Rate	0.2
Lookback Window (T)	24 hours
Prediction Horizons (H)	1, 6, 24 hours
Optimizer	Adam ($\text{lr} = 1 \times 10^{-3}$)
Batch Size	64
Max Epochs	200
Early Stopping Patience	20